Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) who ships full-stack from LLM pipeline to backend to UX. Builds trustworthy AI agents with RAG, context engineering, evals, and guardrails over messy enterprise data, and operates coding agents like Claude Code and Cursor daily. Production experience in Python, TypeScript, and cloud.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, Java, C/C++, SQL, Go.

AI Agents: RAG, context engineering, LLM tool-calling, MCP, evals, guardrails, Gemini, Vertex AI, LangChain.

Web & Backend: React, Node.js, Express, FastAPI, Django, Flask, REST APIs, PostgreSQL, MySQL, Redis.

Developer Tools: Claude Code, Cursor, Git, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, Terraform, Linux, Bash, SonarQube.

Data & ML: PySpark, Delta Lake, ETL pipelines, PyTorch, TensorFlow, scikit-learn, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM agent that decouples context extraction from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Built a config-driven, replay-safe harness with dry-run validation and a versioned run-artifact schema, demonstrating **25.7x latency reduction** (39.4s vs 1012.1s per run) with fidelity preserved.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows in Python (Boto3), replacing legacy Ruby scripts and cutting deployment time by **40%** across AWS environments.
- Executed **44 production upgrades** in 4 months, digging into live issues across Linux, EC2, and databases until root cause was understood.
- Designed **Ansible Blue-Green deployment** with idempotent, audit-ready rollback, recovering any environment in 15 minutes and cutting upgrade downtime **50%**; migrated **40+ environments** to Kubernetes with ArgoCD, Helm, and GitLab CI/CD.
- Served as Quality Coach and SonarQube Specialist, raising automated code coverage from **35% to 78%** and mentoring **5 engineers** through structured onboarding.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Built **Go + Terratest** verification framework that caught **95%** of misconfigurations before they reached production.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and shipped a full-stack **React and MySQL** workflow app that digitized departmental record management and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

PolicyPilot – Policy Intelligence Portal [React, TS, Node, Gemini, MCP, Zod] May 2026

- Architected an evidence-grounded **RAG pipeline** over Gemini 2.5-flash that forces citation-backed answers, so retrieval mismatches surface instead of silently misleading a reviewer on high-stakes eligibility decisions.
- Built a **17-tool MCP server** for structured LLM tool-calling plus a confidence-scoring system that escalates uncertain cases to human review instead of guessing; validated every data boundary with Zod schemas.

Spotify Personal Analytics Platform [Python, FastAPI, PostgreSQL, React, ETL] Jul 2026

- Engineered a 4-stage Python **ETL pipeline** ingesting, normalizing, and reconciling **71K+** events from multiple external APIs with rate limiting, backoff retries, and checkpointing into one PostgreSQL source of truth.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants) | **Published research**, IJRASET | **SAFe Practitioner 5.1**.